

Data Warehouses and Data Mining Systems

Report no. 1

Exercise topic:

Phase 1: Defining a problem

Project topic:

Counter Strike skin market analysis

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1 Topic

The purpose of this research is to analyze the market of weapon skins in the game Counter-Strike. Weapon skins are cosmetic items that can be traded and sold on online marketplaces. Their prices fluctuate depending on various factors such as rarity, condition, popularity of the weapon, and changes in the game's meta.

The goal of this project is to analyze historical price data and identify factors that influence skin prices. The research also aims to determine which skins may be good investment opportunities and how price trends change depending on the popularity of weapons used in the current game meta.

The main research questions are:

- What factors influence the price of counter strike skins?
- Which skins may be worth investing in?
- How does the in-game meta influence the prices of skins?

2 Datasets

The datasets used in this project come from publicly available sources related to the Steam Market and CS:GO trading statistics.

Dataset links:

- <https://www.kaggle.com/datasets/corymusial/cs2-skins-database>
- <https://www.kaggle.com/datasets/kieranpoc/counter-strike-market-sale-data>
- <https://github.com/hojlund123/csgo-dataset>

The dataset consists of two main files:

- **Skin information file** – contains general information about skins such as weapon type, rarity, exterior condition, and collection.
- **Price history file** – contains historical price data collected from the Steam Community Market.
- **Professional matches** – contains historical results of professional e-sport matches.

Important columns include:

- **name** – the name of the skin.
- **weapon** – the weapon type (e.g., AK-47, AWP, M4A4).
- **rarity** – rarity tier of the skin (e.g., Consumer Grade, Covert).
- **wear** – condition of the skin (Factory New, Minimal Wear, Field-Tested, Well-Worn, Battle-Scarred).
- **collection** – the case or collection where the skin originates.
- **timestamp** – the date and time when the price was recorded.
- **price** – the market price of the skin at a given time.
- **volume** – number of transactions or sales.

These attributes allow the analysis of relationships between skin characteristics and market prices.

3 Analysis

The main objective of the analysis is to extract knowledge about price formation and investment potential in the CS skin market.

The analysis will focus on the following aspects:

- **Price determinants** – investigating how factors such as rarity, skin condition, and weapon popularity affect the price of skins.
- **Investment potential** – identifying skins with stable long-term price growth and analyzing patterns that may indicate good investment opportunities.
- **Meta influence** – studying how changes in the in-game meta (for example increased usage of certain weapons in professional play) influence the demand and prices of skins.
- **Price trends over time** – analyzing how prices change across months and years and identifying seasonal patterns.

The expected results of this analysis include discovering correlations between weapon popularity and skin prices, identifying skins with strong market demand, and determining which characteristics make skins more valuable for collectors and traders.